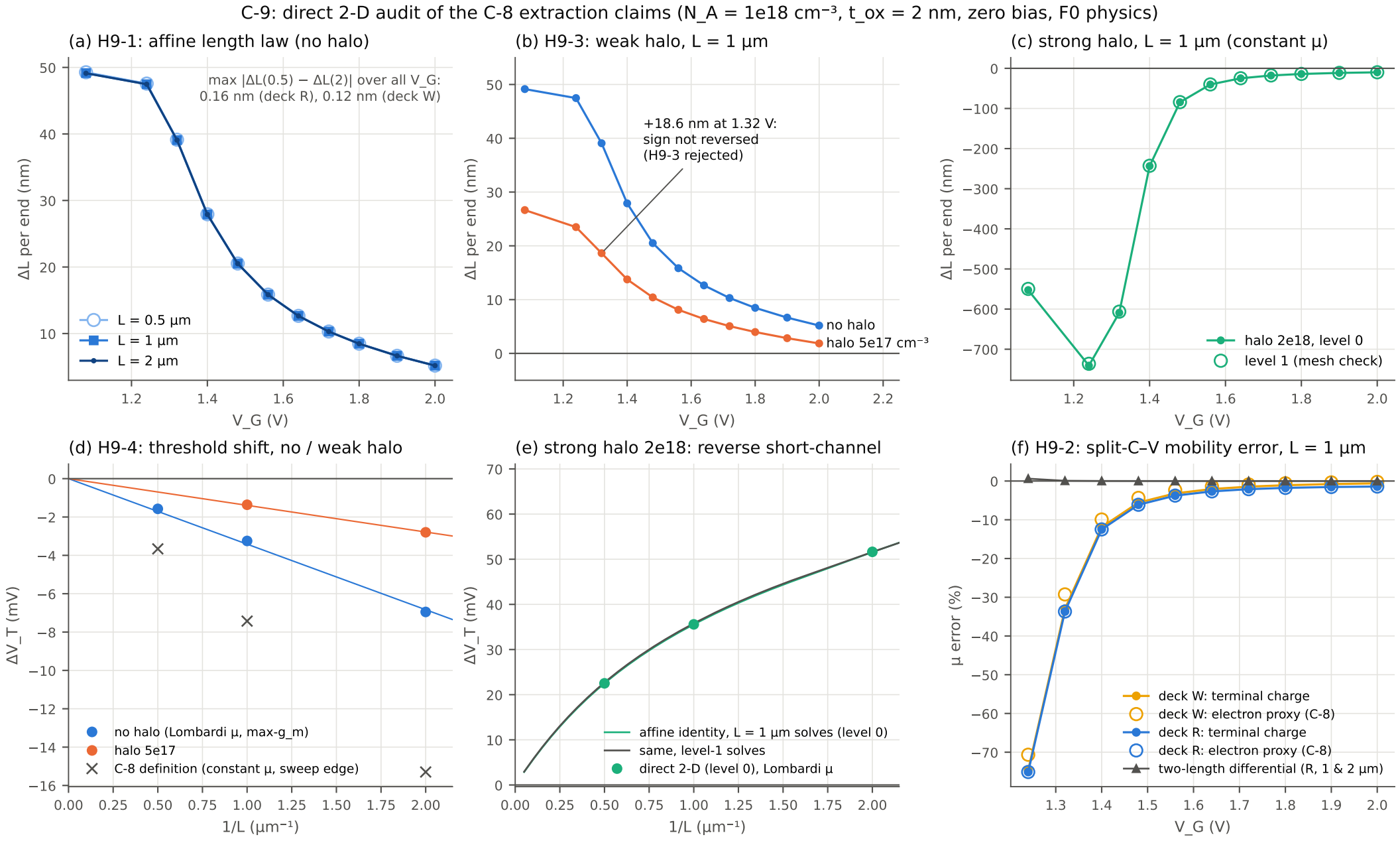
C-9: direct 2-D audit of the C-8 extraction claims (N_A = 1e18 cm⁻³, t_ox = 2 nm, zero bias, F0 physics)
(a) H9-1: affine length law (no halo)
max |ΔL(0.5) − ΔL(2)| over all V_G:
0.16 nm (deck R), 0.12 nm (deck W)
L = 0.5 µm
L = 1 µm
L = 2 µm
ΔL per end (nm)
V_G (V)
(b) H9-3: weak halo, L = 1 µm
+18.6 nm at 1.32 V:
sign not reversed
(H9-3 rejected)
no halo
halo 5e17 cm⁻³
ΔL per end (nm)
V_G (V)
(c) strong halo, L = 1 µm (constant µ)
halo 2e18, level 0
level 1 (mesh check)
ΔL per end (nm)
V_G (V)
(d) H9-4: threshold shift, no / weak halo
no halo (Lombardi µ, max-g_m)
halo 5e17
C-8 definition (constant µ, sweep edge)
ΔV_T (mV)
1/L (µm⁻¹)
(e) strong halo 2e18: reverse short-channel
affine identity, L = 1 µm solves (level 0)
same, level-1 solves
direct 2-D (level 0), Lombardi µ
ΔV_T (mV)
1/L (µm⁻¹)
(f) H9-2: split-C–V mobility error, L = 1 µm
deck W: terminal charge
deck W: electron proxy (C-8)
deck R: terminal charge
deck R: electron proxy (C-8)
two-length differential (R, 1 & 2 µm)
µ error (%)
V_G (V)